Veerapetch Petchger

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EDUCATION

Columbia University, Mailman School of Public Health — New York, NY
M.S. Biostatistics, GPA 3.58 — Expected May 2027

University of Maryland, College Park — College Park, MD
B.S. Public Health Science, Minor in Statistics

EXPERIENCE

Columbia University, Department of Biostatistics — New York, NY
Teaching Assistant, SQL (P8180) — Sep 2026 – Present

- Strengthen the SQL skills of 38 graduate students working with real-world health data by assisting weekly class sessions and checking their queries for correctness.
- Deliver timely, consistent feedback by grading checkpoint design plans and quizzes and resolving grading questions with the instructor.

Columbia University, Department of Biostatistics — Remote
Clinical Research Data Assistant — Aug 2026 – Present

- Advanced the Retired NFL Players Baseline Assessment Program's brain health research by completing 60 case report forms in two months, each drawn from up to 20 medical and neuropsychological reports.
- Safeguard data quality through double-entry verification, with every discrepancy between two independent entries reconciled before analysis.

Columbia University, Mailman School of Public Health — New York, NY
Graduate Research Assistant — Jan 2026 – Present

- Enabled the Pei Lab's analysis of NYC's January 2025 congestion pricing by engineering a 15.8M-row daily ZIP-code mobility dataset from 68 weekly Advan/SafeGraph files in Python.
- Validated a Python rebuild of the lab's R mobility method by reproducing the expected holiday travel dip, implementing tract-to-ZIP mapping, visitor correction, and daily allocation.
- Measured travel shifts across 7 business categories and 5 boroughs by comparing 30 post-policy days with a 13-day baseline in heatmaps.

University of Maryland, School of Public Health — College Park, MD
Graduate Research Assistant — Jul 2024 – Jul 2025

- Established the research protocol for a federally funded, multi-agency pandemic communication study by authoring its IRB exemption proposal.

PROJECTS

South Korea Population Simulator — Aug 2026

- Forecast a century of population decline and rising dependency ratios under South Korea's record-low 0.75 fertility rate in a public cohort-component tool.
- Decomposed Korea's taxable wage base into workforce-decline and wage-growth effects using age–gender wage profiles calibrated to OECD data.

Predicting Severe COVID-19 Infection — May 2026

- Achieved a test AUC of 0.882 on 1,000 participants by benchmarking six classifiers with 10-fold cross-validation.
- Pinpointed vaccination status and systolic blood pressure as the top 2 risk factors, consistent across two models, using variable importance and partial dependence.

Breast Cancer Progression-Free Survival Across Two Cohorts — Mar 2026

- Delivered an interpretable LASSO model for 2,000 patients (test RMSE 0.684) by benchmarking six regression models with 10-fold cross-validation.
- Exposed a 21% rise in prediction error (RMSE 0.684 to 0.829) in a cohort treated a decade later and traced it to shifts in HER2 and hormonal therapy effects, underscoring the need to revalidate models.

SKILLS

Programming: R (tidyverse, caret, tidymodels, glmnet, mgcv, gtsummary), SQL (MySQL, SQLite, H2), SAS, Python (foundational)
Methods: Penalized regression (LASSO, ridge, elastic net), GAM, MARS, random forests, boosting, SVMs, survival analysis, cross-validation, ROC/AUC
Tools: RStudio, R Markdown, Git/GitHub, DbVisualizer, Google Colab, Jupyter
Languages: English, Thai, French

LEADERSHIP

R.I.S.E. Peer Mentor, Columbia Mailman (2026 – Present); Student Ambassador, Department of Biostatistics (2026 – Present)
Co-Founder & Co-President, Thai Student Association, UMD (2022 – 2024); ATSA Southern Region Committee (2023 – 2024)